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Quick Hitter Checklist

- 1) Know your data compression ratio to know your capacity needs
 - a. What is your Compression Ratio (CR)? (measured via Comprestimator, guessed, or hoped for, or seen on the GUI). For example, CR 2:1
 - b. What Capacity (CAP) do you need to present to the host (or SVC cluster)?
 - c. Physical capacity you need is: $(CAP/CR)/.85$
- 2) Activate Call Home
- 3) Monitor (via GUI, or other software)
 - a. Validate that your actual usage matches your plan

How to manage capacity of IBM FlashSystem AE3 - Best Practices

Overview

IBM FlashSystem 900 model AE3 supports in-line, always-on compression. It behaves as a thin-provisioned system. That means that special considerations must be made when provisioning, managing, and monitoring the storage capacity. This document describes the most crucial factors and recommended best practices for provisioning and managing IBM FlashSystem 900 model AE3.

There are three extremely important considerations when managing an AE3 system

1. **The AE3 must be provisioned correctly, based on the compressibility of the actual data.** The Comprestimator tool can be used to determine this before data is stored on the system.
2. **The AE3 must be monitored directly.** The AE3 GUI or Storage Insights must be used to monitor *usable (physical)* capacity used. Email notifications and Call Home should be enabled on the AE3. SVC or other storage virtualization products *do not report the used physical capacity of the AE3 and will not generate notifications or alerts when space is low.*
3. **The correct firmware should be used.** Ensure that the correct, supported firmware levels are used, especially on the host systems and, if virtualized, the storage front end systems, such as IBM San Volume Controller (SVC) or IBM V7000. The recommended level for Storwize products that virtualize storage on a FlashSystem AE3 at the time of this writing is 8.1.1.2. A higher level may be required. See: <http://www-01.ibm.com/support/docview.wss?uid=ssg1S1009645>. Older versions of firmware do not support the SCSI unmap command. If you use the AE3 without unmap support, there is a considerable risk of wasted storage, running out of space, and potential serious problems recovering if an out of space condition occurs.

AE3 Compression Overview

IBM FlashSystem 900 compression algorithm is a Modified Dynamic GZIP algorithm

- Always on. There is no performance penalty, even for incompressible data.
- Combines LZ1 with a form of Pseudo Dynamic Huffman coding.
- Implemented completely in hardware - no processor intervention.
- Takes advantage of already existing LSA (Logical Structured Array) mapping.

Compression and decompression is performed on individual logical pages inside each Flash Module. It is the first step in the inbound data path, before any logical-to-physical mapping occurs. This means there is less data to transfer to the back end making up for small added latency (7uS). Incompressible is detected and handled so that it data only consumes 1:1 (e.g., no meta-data bloat due to existing LSA re-use). Compression and decompression are completely transparent above the flash module except for space management. Performance scales linearly with number of engines. (Each flash module has 1 or 2 engines, depending on the type of Module.)

The AE3 has multiple data protection mechanisms. Decompression is performed as the last step in the *outbound* data path. It happens directly prior to returning the requested data, therefore there is no "store and forward." Decompression is also performed as a "check step" in the *inbound* data path. After incoming data is compressed, it is run through the decompressor real-time and checked against original data before committing. Finally, data protection (ECC) is implemented on top of compressed data. This allows garbage collection and other background data transactions to operate on compressed data.

Phases

The best plan of approach to managing capacity occurs in 4 phases. The earlier the phase that you begin planning, the better the outcome. Although it might not be possible to enjoy all 4 phases, efforts should still to visit each phase.

These phases include: Planning, Monitoring, Prevention, and Recovery.

Phase 1: Planning

Understanding FlashSystem AE3 Capacity

Managing IBM FlashSystem 900 model AE3 requires that you understand the different between usable and effective capacity:

Term	Definition
Usable Capacity	Amount of available space to store user data, <i>after</i> RAID and all data reduction technologies applied. In the AE3 UIs, this is referred to as the Physical Capacity.
Effective Capacity	Amount of user data that could be written, assuming maximum effectiveness of RAID and all data reduction technologies applied.

*NOTE: Although the GUI and CLI list "TB" as units, these are tebibyte units, and in this document is referred to as "TiB". Marketing uses the unit TB (terabyte) as it's a bigger number, but when planning its best to use TiB units.

IBM FlashSystem 900 model AE3 can be configured with four flash module sizes. Flash module sizes cannot be mixed within an enclosure. All the modules must be the same size. It can be ordered with 6, 8, 10, or 12 small capacity modules (3.29TiB which is 3.6TB) or in configurations of 8, 10, or 12 modules when ordered with the medium, or large capacity modules. The modules are described by both the "usable" (physical) capacity, and by the "maximum effective capacity" of the modules.

Flash Module Type	Usable Capacity (TB, marketing)	Usable Capacity (TiB)	Average Effective Capacity @ 2:1 Compression (TiB)	Maximum Effective Capacity (TiB) *
Small	3.61	3.29	6.58	10.01
Medium	8.55	7.78	15.56	20.01
Large	18.00	16.38	*18.00	21.99
Extra Large	18.00	16.38	32.76	40.02

*** Effective Capacity is the total amount of user data that *could be* stored on the Module, assuming the compression ratio of the data is at least equal to or higher than the ratio of Effective Capacity to Usable Capacity.**

The definition above is very important to understand. A flash module's maximum effective capacity, and thus the array's total maximum effective capacity **depends on the compressibility of the data stored**. The table below may help to illustrate this concept. This example assumes there is a **fully populated array of 12 flash modules**:

Physical Capacity (TiB)			Effective Capacity (TiB)													
Flash Module Type	Total Usable per module	Total Usable capacity of the array	Actual compression ratio of the stored data from Comprestimator (n:1)													
			1.0		1.2		1.4		1.6		1.8		2.0		2.2	
			100%	85%	100%	85%	100%	85%	100%	85%	100%	85%	100%	85%	100%	85%
Small	3.29	32.70	32.70	27.80	39.24	33.35	45.78	38.91	52.32	44.47	58.86	50.03	65.40	55.59	71.94	61.15
Medium	7.78	77.60	77.60	65.96	93.12	79.15	108.64	92.34	124.16	105.54	139.68	118.73	155.20	131.92	170.72	145.11
Large	16.38	163.60	163.60	139.06	196.32	166.87	199.90	170.00	199.90	170.00	199.90	170.00	199.90	170.00	199.90	170.00
XLarge	16.38	163.60	163.60	139.06	196.32	166.87	229.04	194.68	261.76	222.50	294.48	250.31	327.20	278.12	359.92	305.93

Physical Capacity (TiB)			Effective Capacity (TiB)													
Flash Module Type	Total Usable per module	Total Usable capacity of the array	Actual compression ratio of the stored data from Comprestimator (n:1)													
			2.4		2.6		2.8		3.0		3.2		3.4		3.6	
			100%	85%	100%	85%	100%	85%	100%	85%	100%	85%	100%	85%	100%	85%
Small	3.29	32.70	78.48	66.71	85.02	72.27	91.56	77.83	98.10	83.39	99.90	84.92	99.90	84.92	99.90	84.92
Medium	7.78	77.60	186.24	158.30	199.90	170.00	199.90	170.00	199.90	170.00	199.90	170.00	199.90	170.00	199.90	170.00
Large	16.38	163.60	199.90	170.00	199.90	170.00	199.90	170.00	199.90	170.00	199.90	170.00	199.90	170.00	199.90	170.00
XLarge	16.38	163.60	392.64	333.74	399.90	339.92	399.90	340.00	399.90	340.00	399.90	340.00	399.90	340.00	399.90	340.00

From the example above, you can see that the amount of *actual* effective capacity depends on the compressibility of the data. When provisioning vdisks in the system, you should not provision more capacity than shown above, for the compression ratio of your data. Notice also that the actual effective capacity cannot be more than the maximum effective capacity for the array (the cells in grey).

Furthermore, for FlashSystem 900, the recommendation for performance is to use no more than 85% of the physical space. There are other considerations regarding reserved space that are covered in the next section. To aid in that calculation, the tables above have the 85% value calculated for the compression ratio of the enclosure capacity.

For more details about IBM FlashSystem 900 AE3, see the Redbook, [Implementing IBM FlashSystem 900 Model AE3](http://www.redbooks.ibm.com/abstracts/sg248414.html?Open)
<http://www.redbooks.ibm.com/abstracts/sg248414.html?Open>

Comprestimator: How well does the data compact?

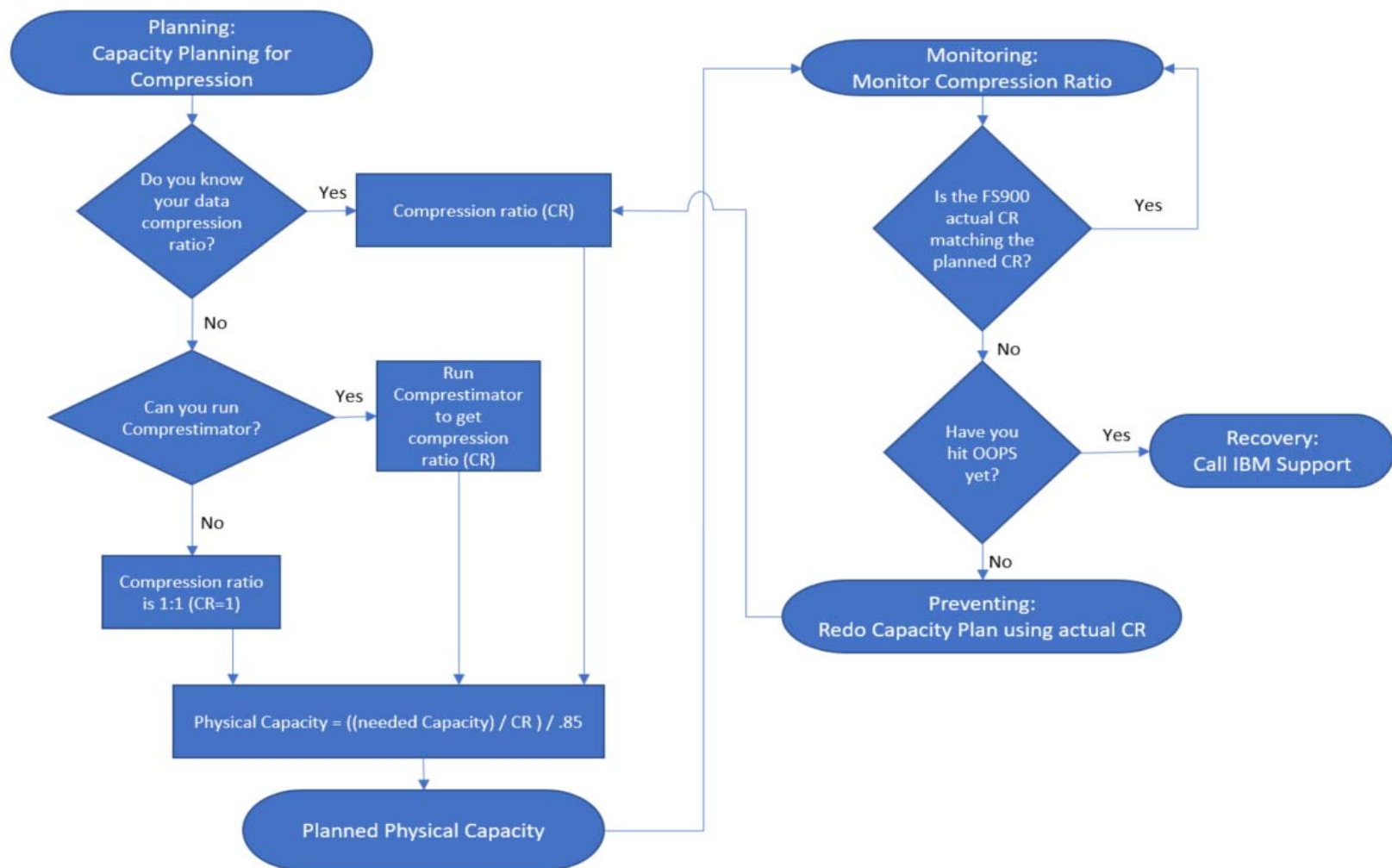
The above information describes the physical and effective capacity, which is very dependent on the nature of the data being written to IBM FlashSystem 900. To best plan the capacity of your data, use the Comprestimator tool against the actual data you intend to put in IBM FlashSystem 900.

Unable to estimate compression? Provision Conservatively

There may be cases where you do not have the ability to estimate or sample the data and so you cannot estimate how compressible it is. In those cases, the best way to safely advance from the planning phase into the deployment and monitoring phase is to assume the worst: assume that the data is completely non-compressible and has a ratio of 1:1. In this extreme case, only provision to 85% of the physical capacity. For example, in table above, the 12 small modules provide just 32.7TiB of physical capacity, which is then also listed as the 32.7TiB effective with a ratio of 1:1. $32.7 \times .85 = 27.8$ TiB of capacity would be provisioned. You may find that as you onboard data, that the compression yield is better than 1:1 (e.g. perhaps its 1.5:1). That then allows to you use more capacity later yet operate safely and without risk at the beginning of your deployment.

Flowchart for Planning, Monitoring, Prevention, & Recovery

To illustrate how the above flowchart works, consider you think your compression ratio is 1.5:1, and that you're



needed capacity is 90TiB. $((90/1.5) / .85 = 70.58\text{TiB})$, so you should acquire and deploy no less than ~71TiB. After deployment, continue to monitor the compression ratio. If the actual ratio is not matching the planned ratio, you will need to act; be it Preventing an out of physical space, or Recovery from it. Note that if your compression ratio is unknown, or, you are not running version 8.1 or above on your Spectrum Virtualize products, you must use the compression ratio of 1:1 as there is no guarantee of compression ratio and you could result in data loss because of that unsupported combination.

Planning for Easy Tier

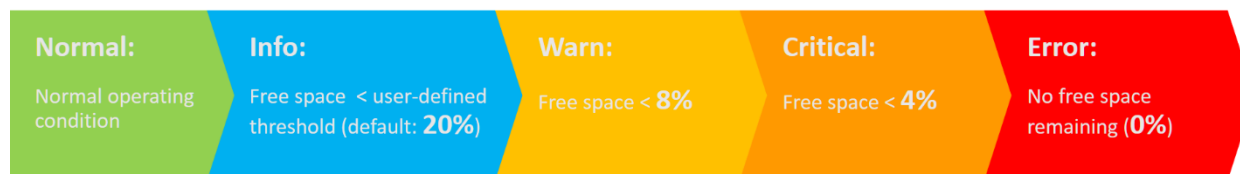
Easy Tier always favors, and will fill up first, the Flash tier. As such, when using FlashSystem in an SVC pool with Easy Tier active, the flash capacity will fill up first, and stay full. Even if the SVC pool has much more lower tier storage, Easy Tier is designed to fill up the top tier first.

Phase 2: Monitoring

Monitoring Capacity

To avoid running out of physical space on the system, the useable capacity should be carefully monitored on the GUI of the AE3. The AE3 GUI is the only capacity dashboard that shows the physical capacity. The virtualization layer (Storwize/SVC/V9000) will only show the effective capacity of the AE3 and will not provide physical space warnings for the AE3 system. Monitoring is especially important when migrating substantial amounts of data onto the AE3, which typically happen during the first part of the workload life cycle as data is onboarded, or initially populated into the storage system.

IBM strongly encourages users to configure Call Home on the AE3. Call Home monitors the physical free space on the system and will automatically open a service call for systems that reach 92% of their useable capacity.



State	Event ID	Level	Event Text	Error Code	Error Code Text (English)
Normal	988010	Info	Array has more space allocated than actual physical space	N/A	Array has more space allocated than actual physical space
Info	988011	Info	Array storage is low on available space	N/A	Array storage is low on available space
Warn	085080	Warn	Array storage is low on available physical space	1480	Array storage is low on space
Critical	085081	Warn	Array storage is critically low on available physical space	1245	Array storage is critically low on space
Error	085082	Error	Hosts are write-protected, array storage is out of space	1048	Unexpected enclosure fault

The GUI will also issue warnings as the system start to run out of capacity. Key, and almost more important than how much data is on IBM FlashSystem 900, is the compression ratio of the data. From the prior Phase 1, if you had planned for a specific compression ratio by using the Comprestimator tool, then monitor the compression ratio on the GUI to verify that is indeed the planned-for ratio. As it's an estimate, it may not be the same. So, monitor the consumption rate as well as the compression ratio.

For example, if you have planned (be it by guessing, or by running Comprestimator) that you will have a 2:1 compression ratio, you may have configured and deployed a 32.7TiB physical config provisioned (following the recommendation of 85% physical) with $(32.7 * .85 * 2)$ 55.59TiB of capacity (be it presented to the hosts, or to an SVC cluster). Periodically monitoring the GUI to see its consumed capacity and the compression ratio allows you to see if you are achieving 2:1 for the compression ratio. There might be some variability of the specific ratio, e.g. 2.01:1 or 1.98:1, which is part of the "actual to plan" monitoring.

Monitoring Software

There are several products and solutions in the market that address storage resource management. IBM offers Spectrum Control, which can monitor for these out of space events as well as capacity utilization. Storage Insight is also available, which is more of a cloud style deployment.

SNMP, emails, and syslog, even REST APIs all provide access to the capacity utilization and these space notification events. Custom integration can also be done by using the SSH CLI and monitoring via running command line interfaces. If you use any of these other methods of monitoring the AE3, make sure you are monitoring the used *physical* capacity, not just the used effective capacity.

Monitoring via CLI

Release 1.6.0.0 and later firmware contains new fields added to the output of the `lsarray` command. When the "`lsarray 0`" command runs, it will produce (along with other useful values) these values of note for monitoring and reporting on compression ratio:

```
physical_free_capacity 3.0TB
physical_capacity 13.2TB
effective_used_capacity 39.1TB
```

The compression ratio is: $\text{effective_used_capacity} / (\text{physical_capacity} - \text{physical_free_capacity})$
For the above example output: $39.1 / (13.2 - 3)$ equates to then 39.1TiB of data effectively being put into 10.2 TiB of underlying physical capacity, which is a compression ratio of 3.83:1.

Phase 3: Prevention

The monitoring phase is important, as the statistics allow for predictions that help then prevent out of space conditions. There are 3 statistics to monitor here:

- Free space: Events are created at 20%, 8% and just 4% remaining.
- Compression Ratio: visible from the GUI
- Overprovision Potential: Computed from the actual compression ratio.

Overprovision Potential is calculated by comparing the compression ratio (i.e. a measure of the exact amount of data compression at the time the measurement is taken) forecasted to the provisioned capacity to that of the underlying physical capacity. In other words, the Overprovision Potential is the amount of space that you *could provision* based on the compression ration of the data that is already stored.

For example, if you have a 32.7TiB physical configuration that you've planned for a 2:1 compression ratio and have initially deployed with 61.15TiB of provisioned volumes (e.g. to hosts, or to an SVC cluster). As the workload begins writing its first (for example) 4TiB of data, if this data then is truly 2:1 compressible, the system will then indicate its effective used capacity is 4TiB of the effective provisioned capacity of 61.15TiB. The 4TiB compresses then down to 2TiB, and the physical used capacity then decreases from 32.7TiB free to 30.7TiB free. The compression ratio then reports 2:1. To forecast the overprovision potential, take the provisioned capacity and divide by the compression ratio: $61.15 / 2 = 30.575\text{TiB}$ which then still then allows all the 30.575TiB of data to fit into the 32.7TiB of physical capacity. **This potential then forecasts an Overprovision Potential that is less than the Physical. In this example, the system is "safe". Use the "Overprovisioned Potential" to predict whether your system is properly over-provisioned.**

To continue that example, let's say the next several TiB of data are written. Examining statistics now shows that of the 9TiB data written has compressed into 6TiB.

- The effective used will report 9TiB
- The physical used will report 6TiB
- And the compression ratio will report as 1.5:1.

The Free space still appears in the "safe" range as $(32.7-6)/32.7 = 81.65\%$ free, well away from a 20% or 8% or 4% threshold. But the Overprovision Potential now becomes $61.15/1.5 = 40.77\text{TiB}$ which is more than the physical capacity of 32.7TiB. This is the first indication of risk as the Overprovision Potential exists. To illustrate why that is a risk, consider then that the server workload could continue to write data at that same compression yield of 1.5:1. The server thinks it has 61.15TiB of total volume space, and it doesn't know any different, so it will then attempt to use all the space that it has apparently been given. It will continue to write data. As that data is stored, if it's still occurring at a 1.5:1 ratio, there is only $32.7 * 1.5 = 49.05\text{TiB}$ that the storage could possibly store. Any of that 1.5:1 data after the 49.05TiB will cause out of space.

To continue to run this example forward in time and continue with the assumption the workload is still operation with 1.5:1 compressible data, the 20% remaining event will trigger when the host writes 39.24TiB, as that will then compress down to 26.16TiB which is 80% of the 32.7TiB physical capacity of the storage array. The 8% space remaining event will trigger once the host writes 45.13TiB, and the 4% event when the host writes 47.09TiB. With the Overprovision Potential being 49.05TiB, there are then 3 warning events (in increasing importance) that occur at 39.24TiB, 45.13TiB, and 47.09TiB. Depending on how aggressive the write pattern is, this could be weeks of time, or just days or even a few hours. Flash is fast, very fast, and it's completely possible for a workload to write 4TiB of data in just 2 hours; so that last warning of 4% remaining might only be a few hours before the storage is completely out of space.

Preventing an out of space impact at this point means recalculating your storage capacity needs. If the planned for 2:1 was a guess, or wasn't accurate enough, and that 1.5:1 is the true nature of the data. As soon as you have evidence/information that the actual is not to plan, you can calculate the additional space needed, to prevent an impact. In this continued example, if the intent was to provide the workload 61.37TiB, and the data would still compress at 1.5:1, then the underlying physical demand is 40.91TiB. Accounting then for 85% physical, you

should re-plan the capacity to $(40.91 / .85)$ 48.13TiB, which then calculates to a shortage (or a need for) $(48.13 - 32.7)$ 15.43TiB of capacity.

FlashSystem 900 firmware levels and out of space buffer

There is a **very** small amount of reserve space on each flash module. Prior to firmware 1.6.0.0 this is 8GB, and at 1.6.0.0 and later this space has been increased to 16GB. This space is designed mainly to provide just enough operational space so that proper garbage collection and volume deletion can be done from the host or from the SVC cluster. This space is not designed to supply the shortage of space due to poor planning of your capacity needs. It should be enough to allow hosts (or SVC) to flush any “in flight” data that occurs at the out of space read only mode.

Phase 4: Recovery

When FlashSystem 900 reaches that last space condition, *out of space*, the device will drop into a read only state. Assessment of data compression ratio and the re-planned capacity estimation (just covered in the last phase) should be done to determine how much actual outstanding storage demand might exist. This additional capacity will likely need to be prepared and presented to the host so that recovery can begin.

WARNING: Running completely out of space can be a very severe situation. Recovery can be extremely complicated and time-consuming. For this reason, it is imperative that proper planning and monitoring be done to avoid reaching this condition.

Software/Firmware Releases and the importance of SCSI Unmap

FlashSystem 900 model AE3 are supported as virtualized capacity to the Spectrum Virtualize family of products (e.g. V7000, SVC) at version 8.1 and newer. Although earlier versions allow the capacity to be technically virtualized, they do not provide the critically needed support of the SCSI Unmap function. This is vital for freeing enough space in the V7000 or SVC cluster to allow IBM FlashSystem 900 to have enough unmapped space to reclaim during the trim operation.

By design, Spectrum Virtualize will consume storage extends of virtualized storage, and never release them (that is, until version 8 and SCSI Unmap). This complicates the usage of IBM FlashSystem 900 model AE3, as it causes a continual increase of consumption with no mechanism or chance of reclaiming any unused space. In Spectrum Virtualize version 8.1 (or newer) the SCSI unmap will flow from the SVC/V7000 to FlashSystem 900 which allows it to know what is and isn't used. Simply upgrading from an older cluster (e.g. 7.8) software level to 8.1 does not retroactively release any storage that may have been previously deleted. For example, if you have an SVC cluster at version 7.8, and run out of space, deleting an SVC vdisk will not cause any flash space to be reclaimed as the SVC still maintains reference to it. Upgrading that cluster to 8.1 still also doesn't trigger SCSI Unmap. Once you are at 8.1, then when you delete an SVC vdisk, the Unmap will be sent to IBM FlashSystem 900, so that its space can then be released.

Reclaiming and Unlock

Once you've assessed and accounted for storage capacity, the first step is to contact Support who can aid in unlocking the read only mode and restoring the operation. The reclamation task can take a long time to run, and larger flash arrays will take longer to recover than smaller ones.

SVC and out of space

It's popular for SVC clusters to virtualize FlashSystem 900, which then means if capacity hasn't been planned for properly, the SVC could have impacts with the mdisks that come from IBM FlashSystem 900 that runs out of space.

There are several possible ways to recover from this condition, which depends on the SVC configuration and any other available space.

SVC: Deleting unneeded vdisks

If possible, within the pool (managed disk group) backed by IBM FlashSystem 900, delete any unnecessary vdisks. Be sure you are at version 8.1 or above or deleting vdisks will have no space reclamation benefits what so ever.

SVC: Migrating vdisks

If possible, moving vdisks from the flash pool to another pool can free up space in the flash pool to allow for space reclamation on the flash enclosure. As this vdisk moves into the new pool, its previously occupied flash extends will be freed up (via SCSI unmap), which then goes to provide more free space to the flash enclosure to be configured to a proper provisioning to support the compression ratio.

SVC: Removing mdisks

In some cases, it's possible to remove mdisks from the pool to force free space to be returned to the flash enclosure. For example, if the deployment of flash behind SVC created sixteen 2.5TiB volumes, then there would be a flash pool on the SVC cluster of size (2.5 x 16) 40TiB. When the SVC removes an mdisk from a pool, it begins evacuating any existing data from that mdisk to the remainder mdisks in the pool. Once the SVC has moved all the data off the mdisk, it is removed from the pool, and can then be unmapped from IBM FlashSystem 900 and deleted. In this case, IBM FlashSystem 900 would return that freed up volume to the free space and might be enough to allow unlocking and restore operation. Remember that the pool must have enough free space to store the data that is being migrated off the mdisk, and IBM FlashSystem 900 must have enough *free* physical space to store all the data that is being migrated into the remaining mdisks in the pool.

This method can be repeated as many times as to provide enough operational free space for IBM FlashSystem 900. Keep in mind that it's not 2.5TiB of space, as the SVC sees the effective capacity before compression, and that to calculate how much physical space that equates, to you must factor in the compression ratio. If IBM FlashSystem 900 is reporting a ratio of 1.5:1, then 2.5TiB represents 1.66TiB of underlying physical capacity that can be reclaimed. So, if your goal was to free up 5TiB of physical, you would need to remove three 2.5TiB mdisks, as 7.5TiB at 1.5:1 ratio is then 5TiB of physical space.

FlashSystem 900 firmware levels and out of space buffer

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Summary

All phases (Planning, Monitoring, Preventing, Recovery), have one thing in common: it's all about planning for your compression ratio. Even if you don't have a plan up front, you're forced to do a plan (or re-plan) to recover. And even if Comprestimator couldn't or wasn't used prior, using the compression ratio of the actual data as reported by the device containing it, the compression ratio is the most important value when calculating the capacity plan.